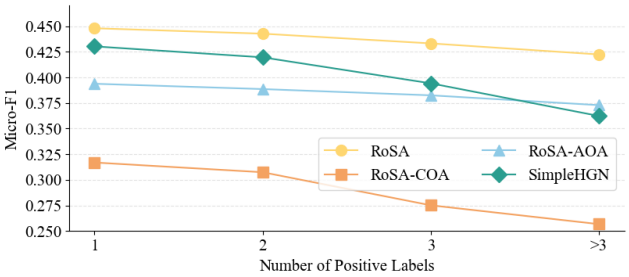

Figure 1. Performance under stratified evaluation by the number of positive labels L.

AOA contribution is defined as:

$$\rho(v) = \frac{\left\| g_v^{(2)} \odot h_{\mathrm{AOA}}(v) \right\|_2}{\left\| g_v^{(1)} \odot h_{\mathrm{COA}}(v) \right\|_2 + \left\| g_v^{(2)} \odot h_{\mathrm{AOA}}(v) \right\|_2} \quad (1)$$

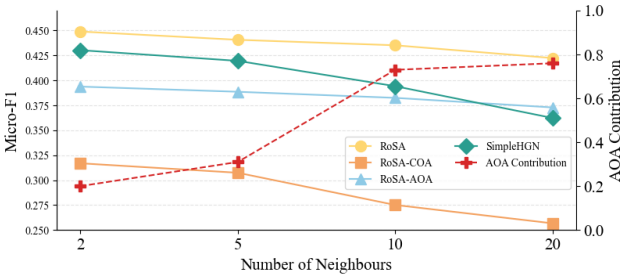

Figure 2. Secondary-neighbor stress test. Model performance: left y-axis; AOA contribution: right y-axis.

Automatic Anchor Selection. We select optimal path P^* from a candidate set $\mathcal{P}_{\mathrm{cand}}$ via

$$P^* = \arg \max_{P \in \mathcal{P}_{\mathrm{cand}}} \frac{1}{2} \left(\text{Micro-F1}_{\mathrm{val}} + \text{Macro-F1}_{\mathrm{val}} \right) \quad (2)$$

Table 1. Performance of anchor selection protocol across datasets

Dataset	Selection	Micro-F1	Macro-F1	Sample-F1
IMDB	All	0.6511	0.6387	0.6420
	Manual	0.6692	0.6477	0.6513
	Automatic	0.6671	0.6415	0.6522
Amazom	All	0.8564	0.7852	0.8331
	Manual	0.9305	0.8184	0.9249
	Automatic	0.9305	0.8184	0.9249
CITE	All	0.3687	0.0095	0.3031
	Manual	0.4224	0.0153	0.4115
	Automatic	0.4006	0.0120	0.3994

Table 2. Performance on IMDB dataset. All models applied the same loss function (except CorGCN and TriPer for they have their own methods). Best results are in **bold**, and second-best are underlined.

Model	Micro-F1	Macro-F1	Sample-F1
RGAT	0.6384	0.5950	0.6228
MAGNN	0.6168	0.5739	0.5918
SimpleHGN	0.6405	0.6084	0.6247
HPN	0.6135	0.5698	0.5890
CorGCN	0.5752	0.1228	0.5171
TriPer	0.6628	0.6462	0.5310
RoSA-COA	0.6277	0.6002	0.6196
RoSA-AOA	0.5708	0.5047	0.5327
RoSA	**0.6692**	**0.6477**	**0.6513**

Table 3. Performance on Amazon dataset. All models applied the same loss function (except CorGCN and TriPer for they have their own methods). Best results are in **bold**, and second-best are underlined.

Model	Micro-F1	Macro-F1	Sample-F1
RGAT	0.7789	0.5362	0.7705
MAGNN	0.9156	0.5631	0.8812
SimpleHGN	0.8843	0.7018	0.8731
HPN	0.6039	0.1816	0.6124
CorGCN	0.5991	0.5487	0.5774
TriPer	0.5660	0.3075	0.5162
RoSA-COA	0.7480	0.6285	0.7797
RoSA-AOA	0.6290	0.1904	0.6256
RoSA	**0.9305**	**0.8184**	**0.9249**

Table 4. Performance on CITE dataset. All models applied the same loss function (except CorGCN and TriPer for they have their own methods). Best results are in **bold**, and second-best are underlined.

Model	Micro-F1	Macro-F1	Sample-F1
RGAT	0.3187	0.0061	0.2746
MAGNN	0.4013	0.0072	0.3971
SimpleHGN	0.3904	0.0069	0.3685
HPN	0.3518	0.0066	0.3441
CorGCN	0.3736	0.0063	0.3706
TriPer	0.1124	0.0070	0.0538
RoSA-COA	0.2567	**0.0255**	0.2484
RoSA-AOA	0.3729	0.0063	0.3712
RoSA	**0.4224**	0.0153	**0.4115**